Lab Report: Training dynamics of deep models

1 Introduction

In this report we compare several models on standard benchmarks. Each experiment was repeated 3 times with different random seeds. The learning rate was set to 0.01 with the Adam optimizer.

2 Results

Figure 1 shows the training loss.

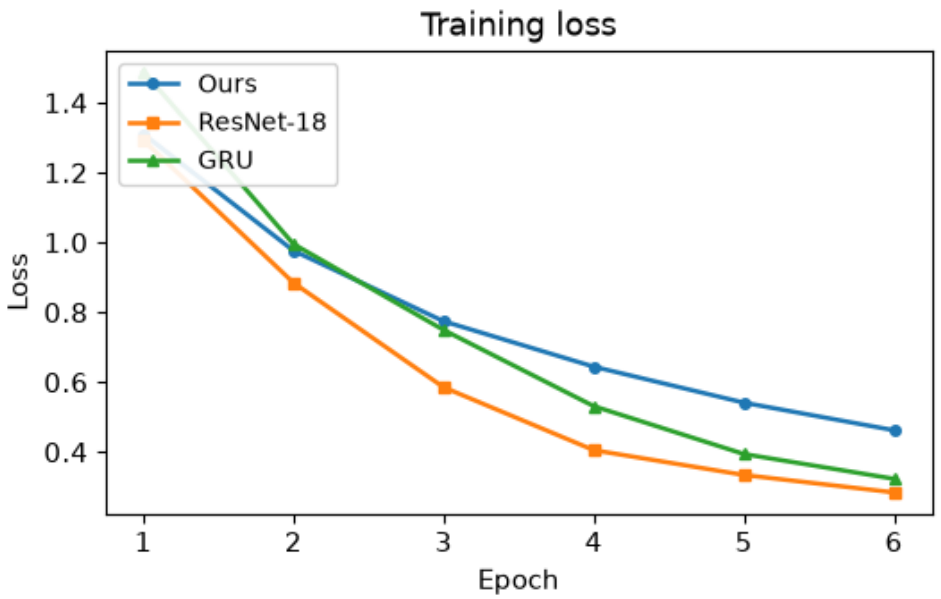


Figure 1: Training loss for the models in this experiment.

As shown in Figure 1, ResNet-18 has a loss of 1.027 at epoch 1. At epoch 3, the loss of Ours is 0.840.

Figure 2 shows the f1 score during training.

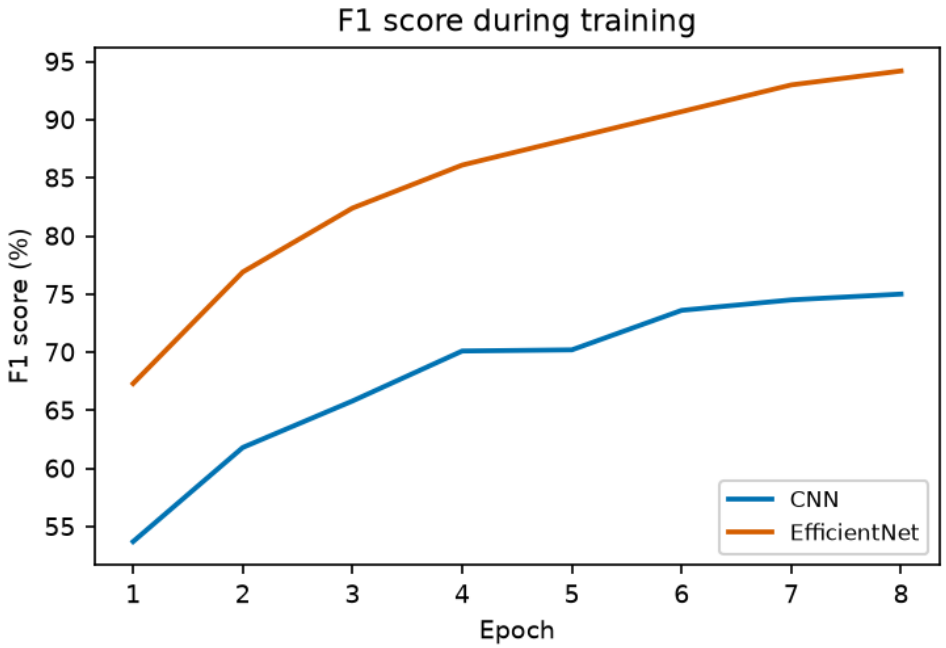


Figure 2: F1 score during training for the models in this experiment.

EfficientNet reaches 93.0% F1 score at epoch 7 (Figure 2). From epoch 1 to epoch 7, the F1 score of EfficientNet improves by a relative 38.2%. CNN attains a F1 score of 57.3% at epoch 2.

Figure 3 shows the f1 score of MLP by dataset.

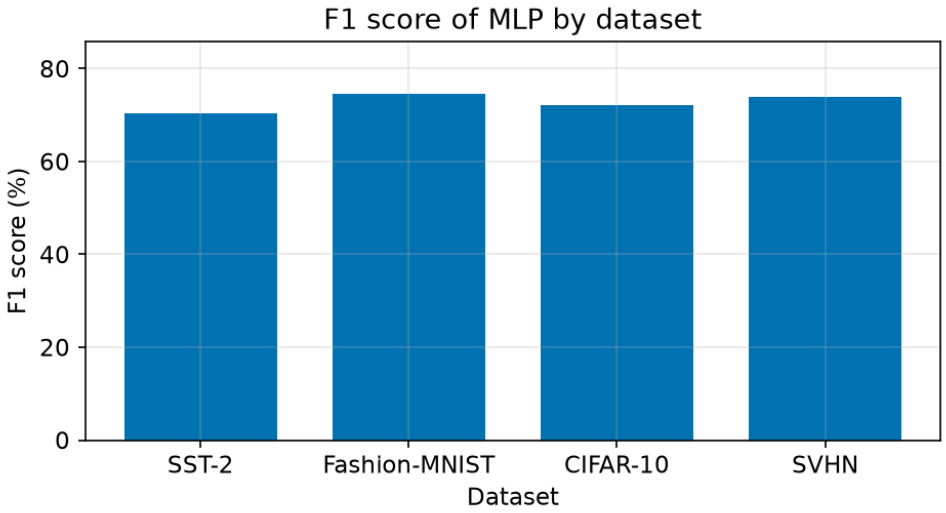


Figure 3: F1 score of MLP by dataset for the models in this experiment.

As shown in Figure 3, MLP reaches 116.9% F1 score on SVHN. On Fashion-MNIST, MLP obtains a F1 score of 75%. MLP reaches 72% F1 score on CIFAR-10.

3 Discussion

Overall, the results are in line with what we expected from the literature.